

TOPIC: REAL NUMBER SYSTEM**GROUP – A (MULTIPLE CHOICE QUESTIONS) [10 × 1 = 10 MARKS]**

1. Which of the following is an irrational number?

- A. $\sqrt{23}$ B. $\sqrt{225}$ C. 0.3799 D. $7.\overline{478}$

2. Which of the following numbers is irrational?

- A. $\sqrt{\frac{4}{9}}$ B. $\frac{\sqrt{1250}}{\sqrt{8}}$ C. $\sqrt{8}$ D. $\frac{\sqrt{24}}{\sqrt{6}}$

3. The product of two irrational numbers is:

- A. always irrational B. always rational C. always an integer D. sometimes rational & sometimes irrational

4. A rational number lying between $\sqrt{2}$ and $\sqrt{3}$ is:

- A. $\frac{\sqrt{2}+\sqrt{3}}{2}$ B. $\sqrt{6}$ C. 1.6 D. 1.9

5. If $x = \frac{\sqrt{7}}{5}$ and $\frac{5}{x} = p\sqrt{7}$, then the value of p is:

- A. $\frac{7}{25}$ B. $\frac{25}{7}$ C. $\frac{7}{15}$ D. $\frac{15}{7}$

6. Which of the following is a true statement?

- A. π and $\frac{22}{7}$ are both rationals.
B. π and $\frac{22}{7}$ are both irrationals.
C. π is rational and $\frac{22}{7}$ is irrational.
D. π is irrational and $\frac{22}{7}$ is rational.

7. If $\sqrt{2} = 1.414$, then the value of $\frac{1}{\sqrt{2}}$ is:

- A. 0.075 B. 0.75 C. 0.705 D. 7.05

8. A rational number between -3 and 3 is:

- A. 0 B. -4.3 C. -3.4 D. 1.101100110001...

9. Every point on a number line represents:

- A. a rational number B. a natural number C. an irrational number D. a unique number

10. Two rational numbers between $\frac{2}{3}$ and $\frac{5}{3}$ are:

A. $\frac{1}{6}$ & $\frac{2}{6}$

B. $\frac{1}{2}$ & $\frac{2}{1}$

C. $\frac{5}{6}$ & $\frac{7}{6}$

D. $\frac{2}{3}$ & $\frac{4}{3}$

GROUP – B [4 × 2 = 8 MARKS]

11. If $x = 9 - 4\sqrt{5}$, find the value of $x^2 + \frac{1}{x^2}$.

12. Prove that:

$$\frac{1}{3 + \sqrt{7}} + \frac{1}{\sqrt{7} + \sqrt{5}} + \frac{1}{\sqrt{5} + \sqrt{3}} + \frac{1}{\sqrt{3} + 1} = 1$$

13. Find two different rational numbers between the rational numbers $\frac{3}{5}$ and $\frac{4}{5}$.

14. Without actual division, find which of the following rational numbers are terminating decimals:

i) $\frac{5}{32}$

ii) $\frac{11}{24}$

iii) $\frac{3}{40}$

GROUP – C [2 × 3 = 6 MARKS]

15. Prove that $\sqrt{3}$ is an irrational number.

16. Represent $\sqrt{5}$ on the real number line.